

Basic Laws

- 1) Ohm's law: It states that the voltage ' V ' across a resistor is directly proportional to the current ' I ' flowing through the resistor.

$$V \propto I$$

Formula -

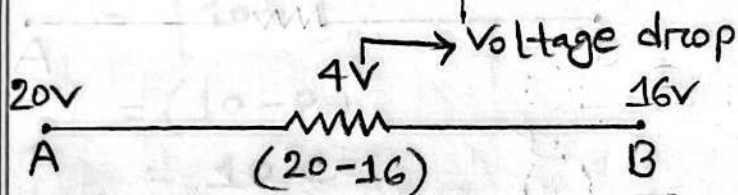
$$V = IR$$

where,

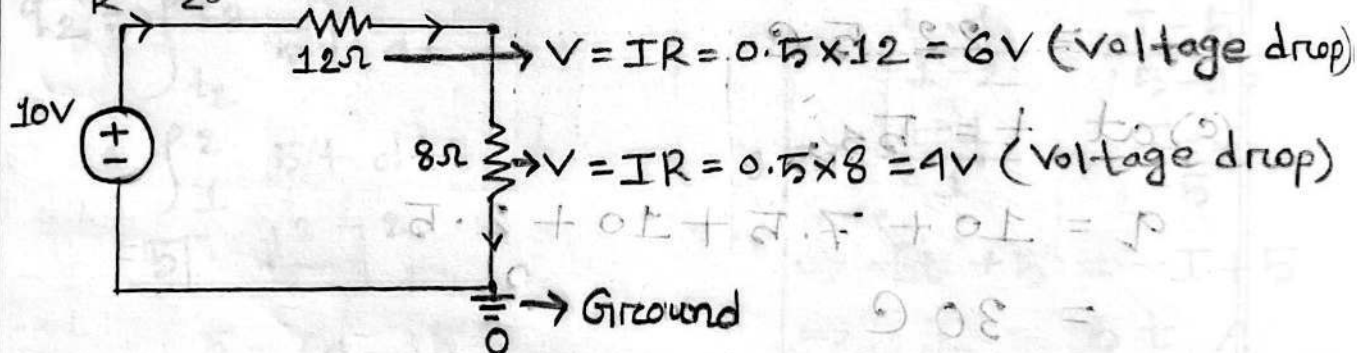
V = voltage drop (V)

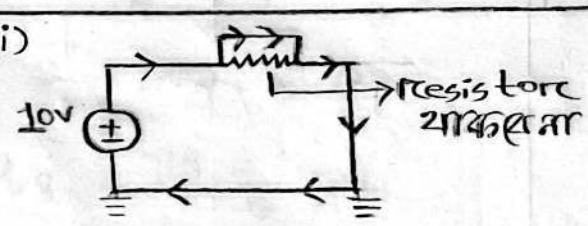
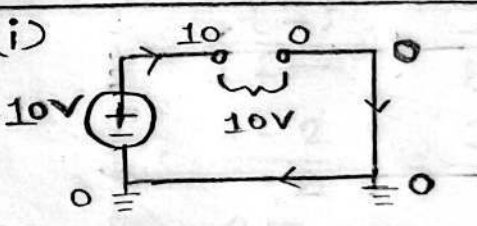
R = resistance (Ω)

I = current (A)

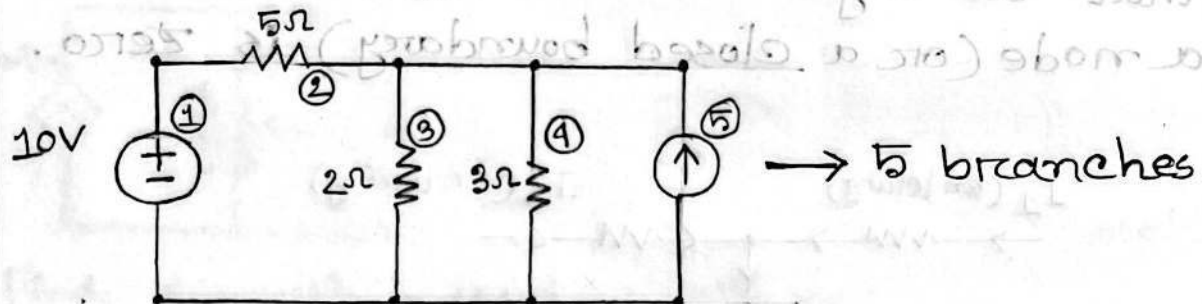


$$I = \frac{V}{R} = \frac{10}{20} = 0.5A$$

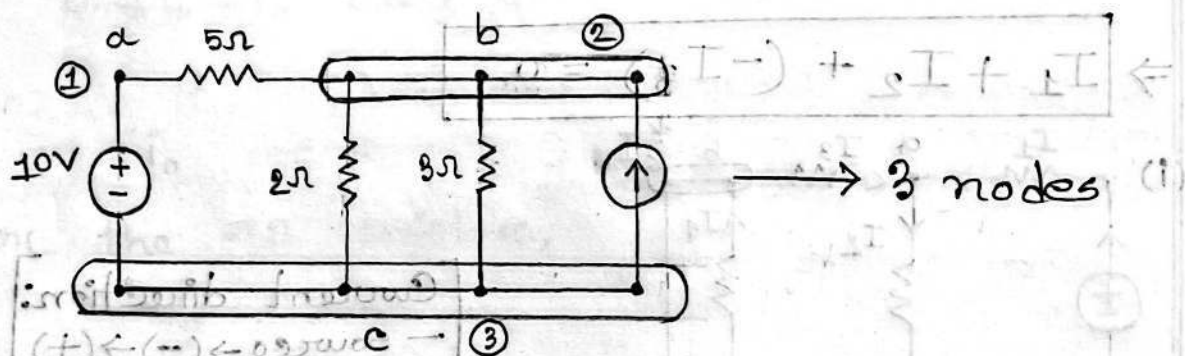


Short Circuit	Open Circuit
(i) 	(i) 
(ii) $R=0$; $V=0$	(ii) $I=0$; $R=\infty$

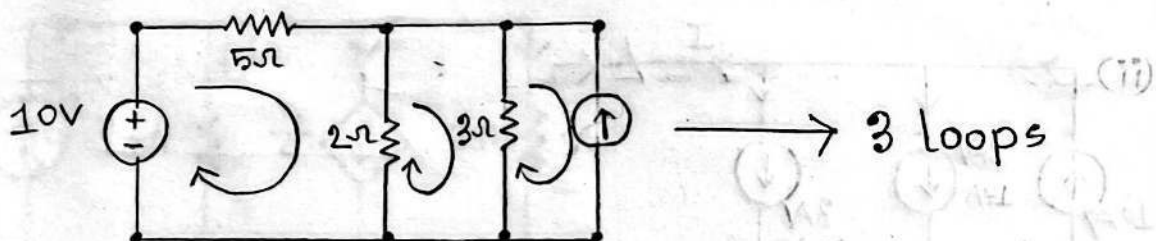
- Branch: A branch represents a single element such a voltage source or a resistor.



- Node: A node is the point of connection between two or more branches.



- Loop: A loop is any closed path in a circuit.



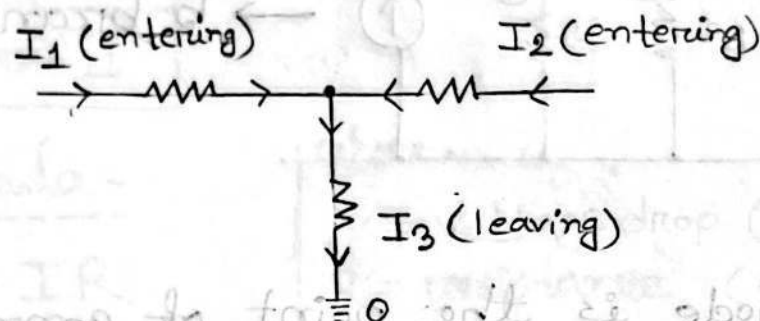
Fundamental theorem -

$$B = N + L - 1$$

$$\Rightarrow 5 = 3 + 3 - 1$$

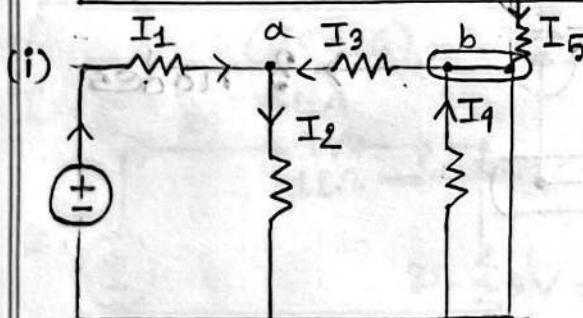
$$\Rightarrow 5 = 5$$

2) Kirchhoff's current law (KCL): It states that the algebraic sum of currents entering a node (or a closed boundary) is zero.



$$I_1 + I_2 = I_3$$

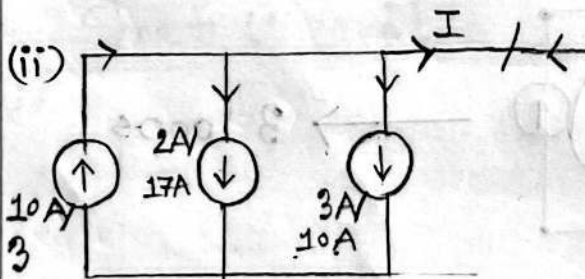
$$\Rightarrow I_1 + I_2 + (-I_3) = 0$$



Current direction:
 - Source $\rightarrow (-) \rightarrow (+)$
 - Ground
 - Resistor $\rightarrow (+) \rightarrow (-)$

(a) $I_1 + I_3 = I_2$

(b) $I_4 + I_5 = I_3$



$$10 = 2 + 3 + I \quad \therefore 3 = I + 17 + 10$$

$$\Rightarrow 5 + I = 10$$

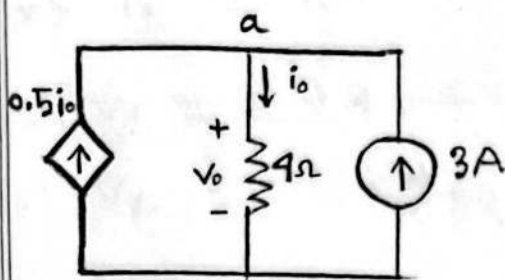
$$\Rightarrow I = 5 \text{ A}$$

$$\Rightarrow 27 + I = 3$$

$$\Rightarrow I = -24 \text{ (leaving)}$$

Example -

2.7]

Find current i_o and voltage v_o :Applying KCL to node a ,

$$0.5i_o + 3A = i_o$$

$$\Rightarrow 0.5i_o - i_o = -3A$$

$$\Rightarrow -0.5i_o = -3A$$

$$\Rightarrow i_o = 6A$$

For the 4Ω resistor,

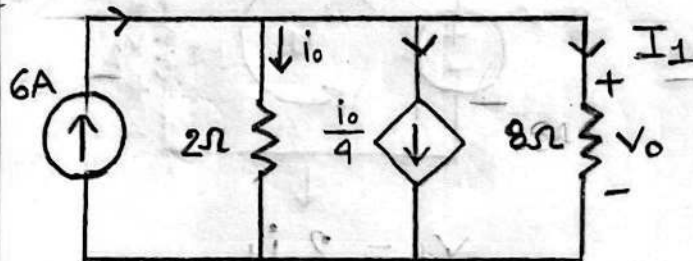
$$v_o = 4 \times i_o$$

$$v_o = (4 \times 6)V = 24V$$

(Ans.)

Mid

2.26]

Find v_o and i_o :

Applying KCL at node 1,

$$6 = i_o + \frac{i_o}{4} + I_1$$

$$\Rightarrow 6 = \frac{v_o}{2} + \frac{v_o}{8} + \frac{v_o}{8}$$

Here,

$$v_o = 2 \times i_o$$

$$\Rightarrow i_o = \frac{v_o}{2}$$

$$v_o = 8 \times I_1$$

$$\Rightarrow I_1 = \frac{v_o}{8}$$

$$\Rightarrow 6 = \frac{4V_o + V_o + V_o}{8}$$

$$\Rightarrow 48 = 6 V_o$$

$$\Rightarrow V_o = 8 \text{ V}$$

Now,

$$i_o = \frac{V_o}{2} = \frac{8}{2} = 4 \text{ A}$$

$$I_1 = \frac{V_o}{8} = \frac{8}{8} = 1 \text{ A}$$

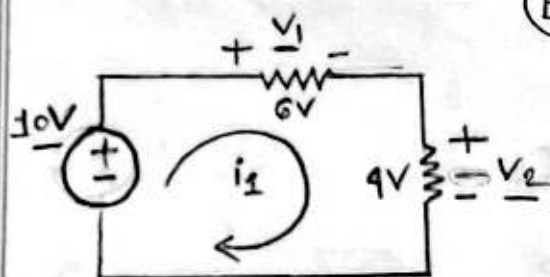
(Ans.)

3] Kirchhoff's Voltage Law (KVL): It states that the algebraic sum of all voltages around a closed path (or loop) is zero.

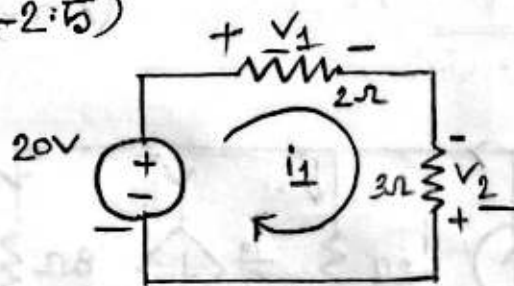
Sum of Voltage supply = Sum of voltage drops

Example -

(Ex-2.5)



KVL,
 $V_1 + V_2 + (-10) = 0$



$$V_1 = 2 i_1$$

$$V_2 = 3 \times (-i_1)$$

$$= -3 i_1$$

KVL,
 $-20 + V_1 - V_2 = 0$

$$\Rightarrow -20 + 2 i_1 - (-3 i_1) = 0$$

$$\Rightarrow -20 + 2 i_1 + 3 i_1 = 0$$

$$\Rightarrow 5i_1 = 20$$

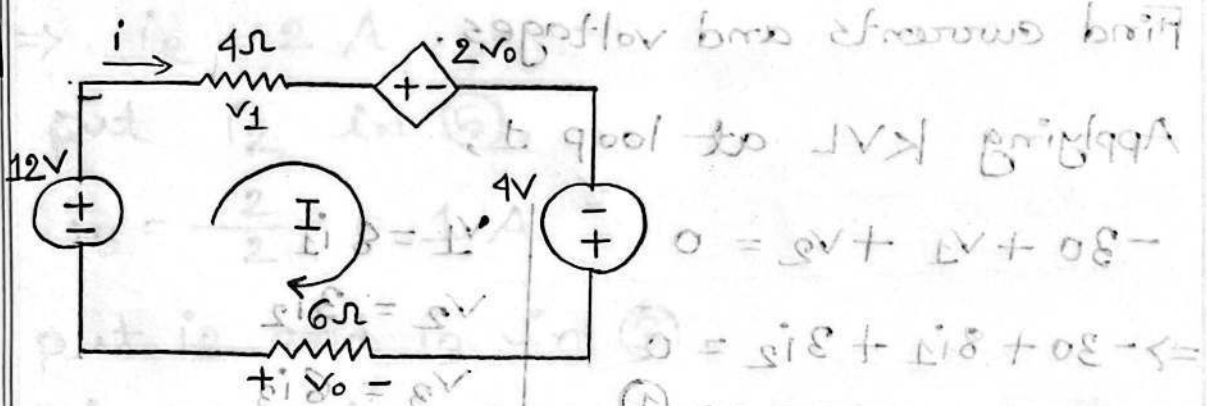
$$\Rightarrow i_1 = 4$$

$$\therefore V_1 = 2 \times 4 = 8 \text{ V}$$

$$\therefore V_2 = -3 \times 4 = -12 \text{ V}$$

(Ans.)

Ex-2.6

Find v_0 and i .

Applying KVL at loop 1,

$$-12 + V_1 + 2V_0 - 4 - V_0 = 0$$

$$\Rightarrow -12 + 4i + 2(-6i) - 4 - (-6i) = 0$$

$$\Rightarrow -12 + 4i - 12i - 4 + 6i = 0$$

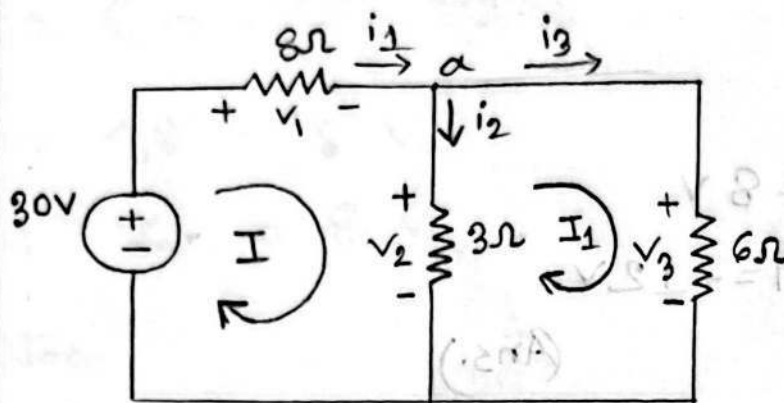
$$\Rightarrow -2i = 16$$

$$\Rightarrow i = -8 \text{ A}$$

$$\therefore i = -8 \text{ A and } V_0 = 48 \text{ V}$$

(Ans.)

2.81



Find currents and voltages.

Applying KVL at loop 1,

$$\begin{array}{l|l} -30 + v_1 + v_2 = 0 & v_1 = 8i_1 \\ \Rightarrow -30 + 8i_1 + 3i_2 = 0 & v_2 = 3i_2 \\ \text{----- ①} & v_3 = 6i_3 \end{array}$$

Applying KVL at loop 2,

$$\begin{array}{l} -v_2 + v_3 = 0 \\ \Rightarrow -3i_2 + 6i_3 = 0 \text{ --- ②} \end{array}$$

Applying KCL at node a,

$$i_1 = i_2 + i_3 \text{ --- ③ } [\because \text{Entering Current} = \text{leaving Current}]$$

① and ③,

$$\begin{array}{l} -30 + 8(i_2 + i_3) + 3i_2 = 0 \\ \Rightarrow 8i_2 + 8i_3 + 3i_2 = 30 \\ \Rightarrow 11i_2 + 8i_3 = 30 \text{ ----- ④} \end{array}$$

$$\textcircled{2}, i_3 = \frac{i_2}{2} \text{ --- } \textcircled{5}$$

put i_3 in $\textcircled{4}$,

$$11i_2 + 8 \times \frac{i_2}{2} = 30$$

$$\Rightarrow 11i_2 + 4i_2 = 30$$

$$\Rightarrow i_2 = 2 \text{ A}$$

put i_2 in $\textcircled{5}$,

$$i_3 = \frac{2}{2} = 1 \text{ A}$$

put i_2 and i_3 in $\textcircled{3}$,

$$i_1 = 2 + 1 = 3 \text{ A}$$

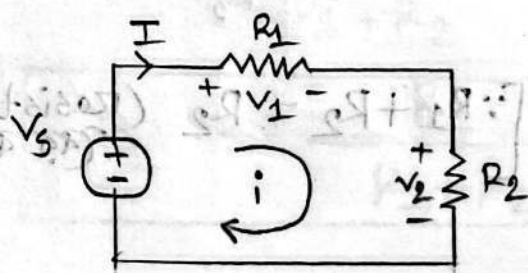
$$\therefore V_1 = 8 \times 3 = 24 \text{ V}$$

$$\therefore V_2 = 3 \times 2 = 6 \text{ V}$$

$$\therefore V_3 = 6 \times 1 = 6 \text{ V}$$

(Ans.)

• Voltage Divider Rule (VDR): (Series)



Applying KVL at loop 1,
 $-V_s + V_1 + V_2 = 0$

$$\begin{aligned} \therefore V_1 &= I R_1 \\ \therefore V_2 &= I R_2 \end{aligned}$$

$$\Rightarrow IR_1 + IR_2 = V_3 \quad \text{--- (3)}$$

$$\Rightarrow I(R_1 + R_2) = V_3$$

$$\Rightarrow I = \frac{V_3}{R_1 + R_2}$$

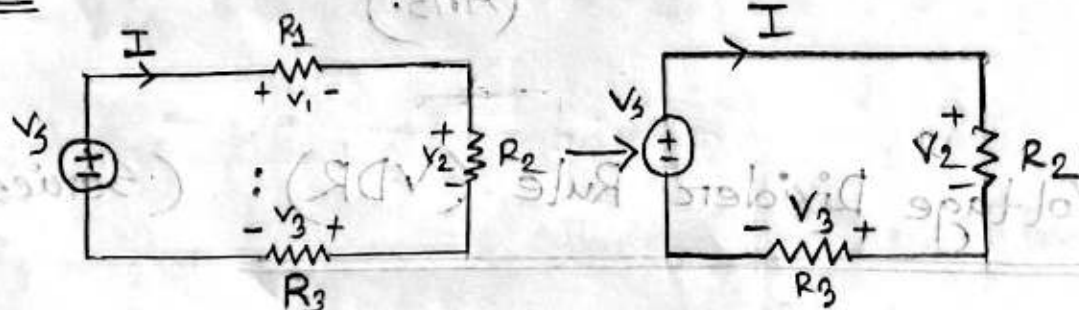
$$\therefore V_1 = \frac{V_3}{R_1 + R_2} \cdot R_1$$

$$\Rightarrow V_1 = \frac{R_1}{R_1 + R_2} \cdot V_3$$

$$\therefore V_2 = \frac{V_3}{R_1 + R_2} \cdot R_2$$

$$\Rightarrow V_2 = \frac{R_2}{R_1 + R_2} \cdot V_3$$

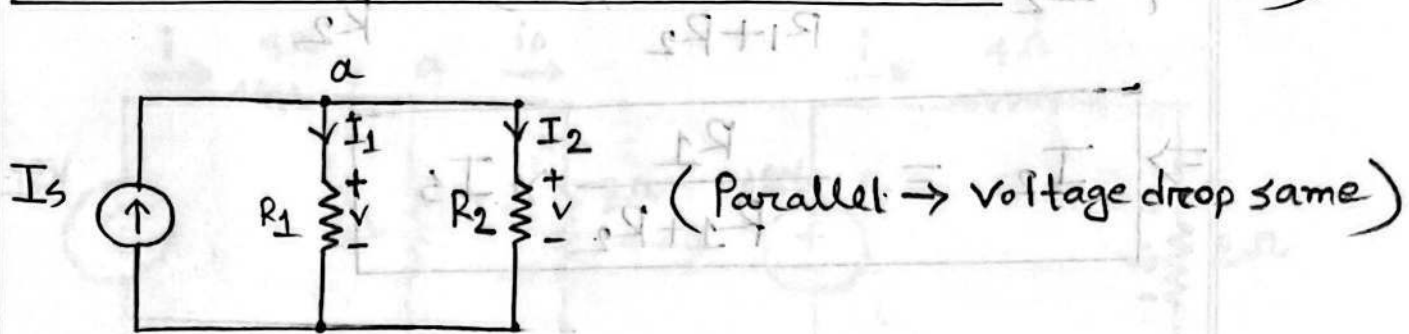
Ex -



$$\therefore V_3 = \frac{R_3}{R_2 + R_3} \cdot V_3$$

$$\therefore R_1 + R_2 = R_2 \quad (\text{resistance})$$

• Current Divider Rule (CDR) : (Parallel)



Applying KCL at node a,

$$I_s = I_1 + I_2$$

$$\Rightarrow I_s = \frac{v}{R_1} + \frac{v}{R_2}$$

$$\Rightarrow I_s = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\therefore v = I_1 R_1$$

$$\therefore v = I_2 R_2$$

$$\Rightarrow I_s = \frac{v(R_1 + R_2)}{R_1 R_2}$$

$$\Rightarrow v = \left(\frac{R_1 R_2}{R_1 + R_2} \right) I_s$$

$$\therefore I_1 = \frac{v}{R_1}$$

$$\Rightarrow I_1 = \frac{R_1 R_2}{R_1 + R_2} \times I_s \times \frac{1}{R_1}$$

$$\Rightarrow I_1 = \frac{R_2}{R_1 + R_2} \times I_s$$

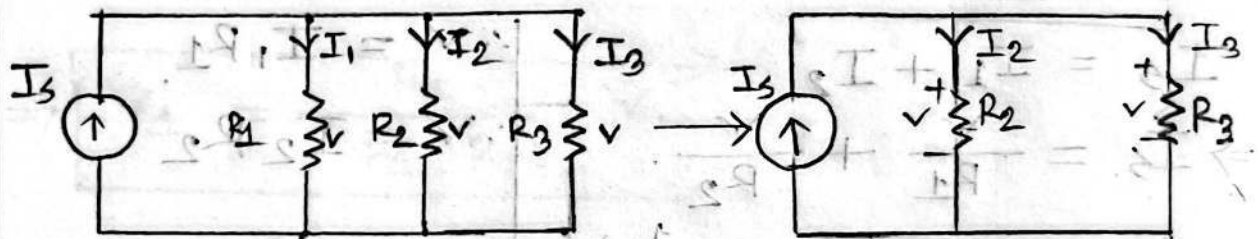
$$\therefore I_2 = \frac{v}{R_2}$$

12.5.21

16.2.26

$$\Rightarrow I_2 = \frac{R_1 R_2}{R_1 + R_2} \times I_s \times \frac{1}{R_2}$$

$$\Rightarrow I_2 = \frac{R_1}{R_1 + R_2} \times I_s$$

Ex -

$$\therefore I_3 = \frac{R_2}{R_2 + R_3} \times I_s$$

(resistance less than)

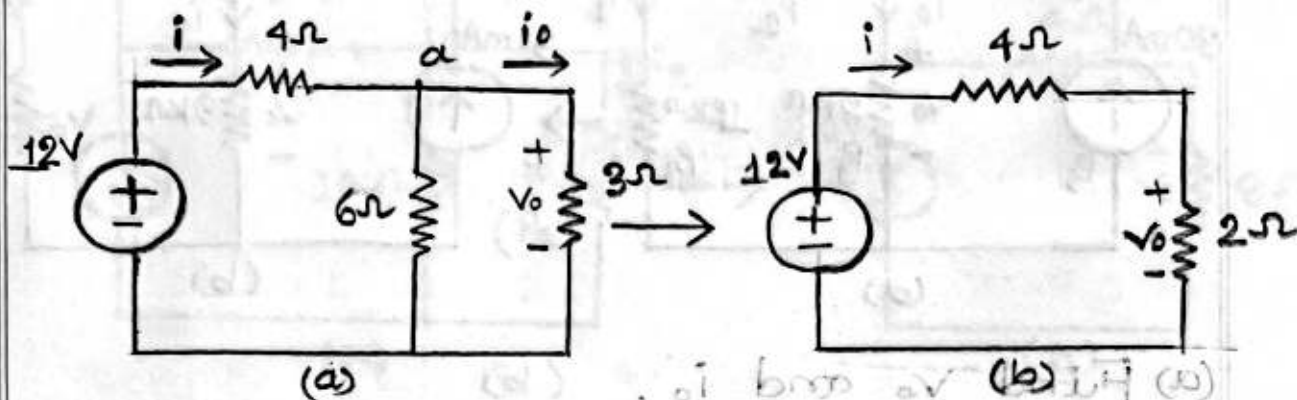
Formula - (OR)

$$I_3 = \frac{R_1 R_2}{R_1 R_2 + R_2 R_3 + R_3 R_1} \times I_s$$

$$I_4 = \frac{R_1 R_2 R_3}{R_1 R_2 R_3 + R_2 R_3 R_4 + R_3 R_4 R_1 + R_4 R_1 R_2} \times I_s$$

Example -

2.12]

Find V_o and i_o .

VDR:

$$R_1 = 6\Omega$$

$$R_2 = 3\Omega$$

$$\therefore \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\Rightarrow \frac{1}{R_p} = \frac{R_2 + R_1}{R_1 R_2}$$

$$\Rightarrow R_p = \frac{R_1 R_2}{R_2 + R_1} = \frac{6 \times 3}{6 + 3} = \frac{18}{9} = 2\Omega$$

$$\therefore V_o = \frac{R_p}{R_1 + R_p} \times V_s = \frac{2}{6 + 2} \times 12 = 4V$$

$$\therefore i_o = \frac{V_o}{R} = \frac{4V}{3\Omega} = \frac{4}{3}A$$

IDR:

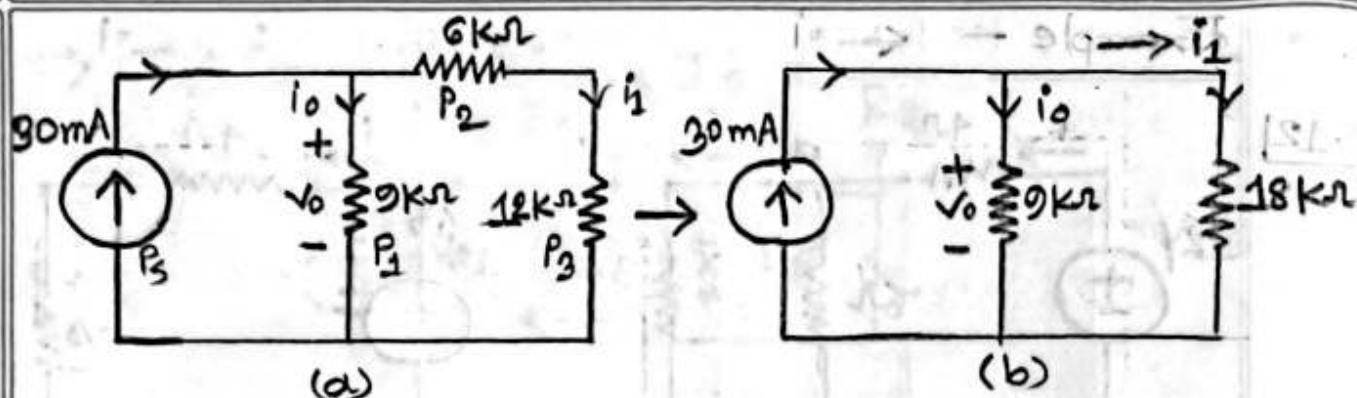
$$i = \frac{V}{R} = \frac{12}{6} = 2A$$

$$i_o = \frac{R_1}{R_1 + R_2} \times i = \frac{6}{6 + 3} \times 2 = \frac{4}{3}A$$

$$V_o = i_o R = \frac{4}{3} \times 3 = 4V \quad (Ans.)$$

2.13

19.2.26

(a) Find V_0 and i_0 . (b)

$$i_0 = \frac{18}{9+18} \times 30 = 20 \text{ mA} \quad \left[\because \frac{K\Omega}{K\Omega} \times \text{mA} = \text{mA} \right]$$

$$V_0 = i_0 \times R_1 = 20 \times 9 \times 10^3 \times 10^3 = 180 \text{ V}$$

(b) Find P_3 .

$$P_3 = -V_0 \times i_s = -180 \times 30 \times 10^{-3} = -5.4 \text{ W (Supply)}$$

(c) Find P_1 , P_2 and P_3 .

$$P_1 = V_0 \times i_0 = 180 \times 20 \times 10^{-3} = 3.6 \text{ W (absorb)}$$

$$P_2 = i_1^2 \times R_2 \quad \left[\because i_1 = \frac{V_0}{R_2} = \frac{180}{18} = 10 \text{ mA} \right] \rightarrow (b)$$

$$= (10 \times 10^{-3})^2 \times 6 \times 10^3 = 0.6 \text{ W (absorb)}$$

$$P_3 = i_1^2 \times R_3$$

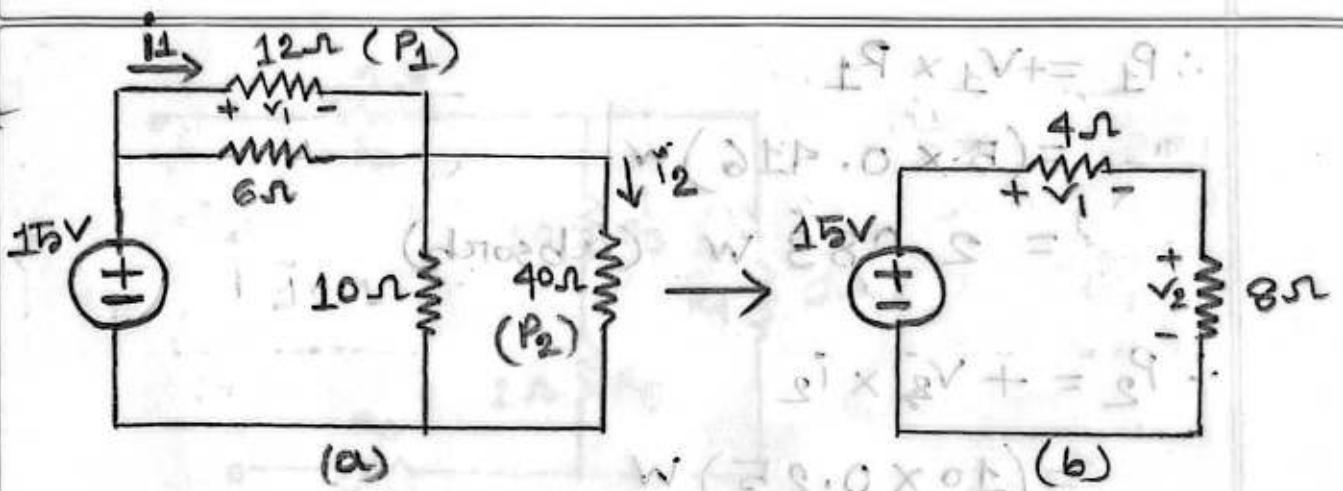
$$= (10 \times 10^{-3})^2 \times 12 \times 10^3$$

$$= 1.2 \text{ W (absorb)} \quad (\text{Ans.})$$

$$\therefore P = VI$$

$$\therefore P = I^2 R \quad [V = IR]$$

$$\therefore P = \frac{V^2}{R} \quad [I = V/R]$$



Find v_1 , v_2 , i_1 , i_2 and P_1 (12Ω), P_2 (40Ω).

$$R_1 = 12\Omega$$

$$R_2 = 6\Omega$$

$$\therefore R_p = \frac{12 \times 6}{12 + 6} = 4\Omega$$

$$R_3 = 10\Omega$$

$$R_4 = 40\Omega$$

$$\therefore R_p = \frac{10 \times 40}{10 + 40} = 8\Omega$$

VDR - (b)

$$\therefore v_1 = \frac{4}{8 + 4} \times 15 = 5V$$

$$\therefore v_2 = \frac{8}{4 + 8} \times 15 = 10V$$

$$\therefore i_1 = \frac{v_1}{R_1} = \frac{5}{12} = 0.416 A = 0.416 \times 10^3 = 416.7 \text{ mA}$$

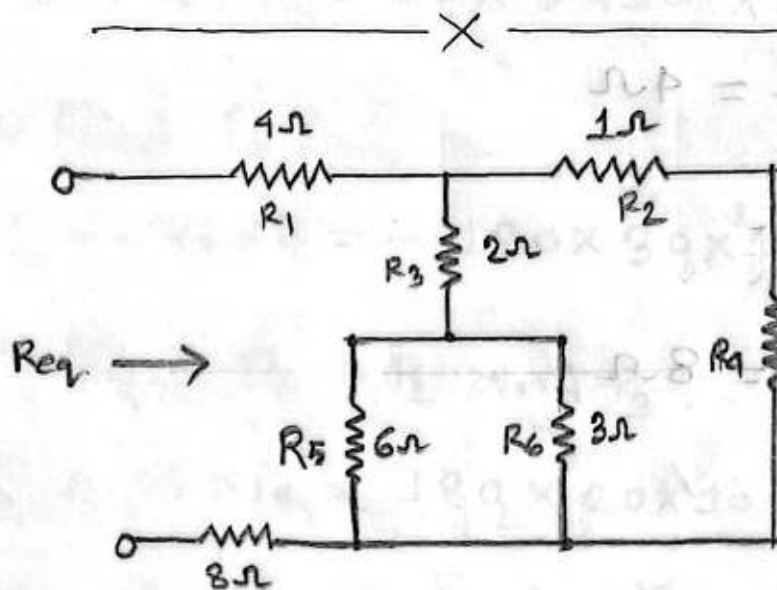
$$\therefore i_2 = \frac{v_2}{R_4} = \frac{10}{40} = 0.25 A = 0.25 \times 10^3 = 250 \text{ mA}$$

$$\begin{aligned}\therefore P_1 &= +V_1 \times I_1 \\ &= (5 \times 0.416) \text{ W} \\ &= 2.083 \text{ W (absorb)}\end{aligned}$$

$$\begin{aligned}\therefore P_2 &= +V_2 \times I_2 \\ &= (10 \times 0.25) \text{ W} \\ &= 2.5 \text{ W (absorb)}\end{aligned}$$

(Ans.)

2.9]

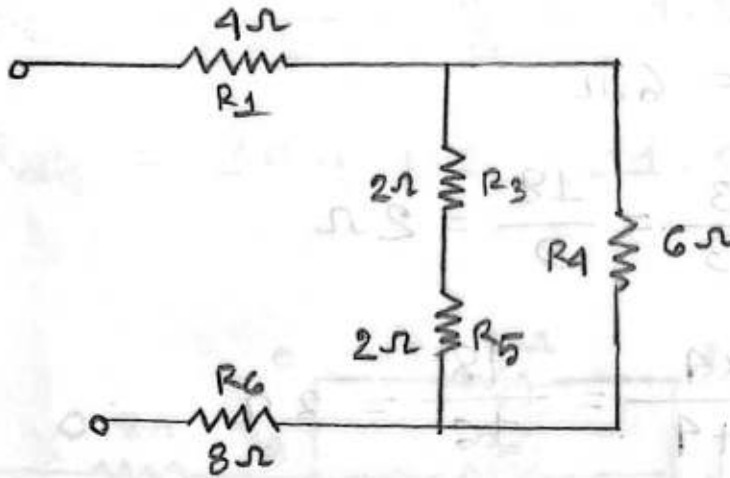


Here,

$$R_2 = R_1 + R_4 = 1 + 5 = 6 \Omega$$

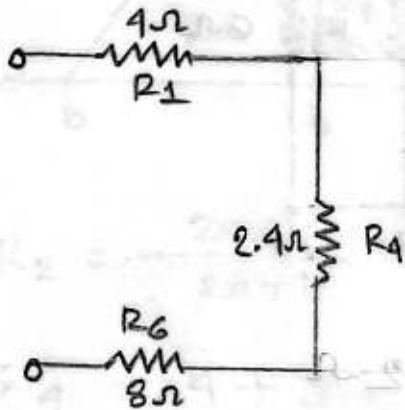
$$\frac{1}{R_5} = \frac{1}{R_6} + \frac{1}{R_5} = \frac{1}{6} + \frac{1}{3} = \frac{6+3}{18}$$

$$\therefore R_5 = \frac{18}{9} = 2 \Omega$$



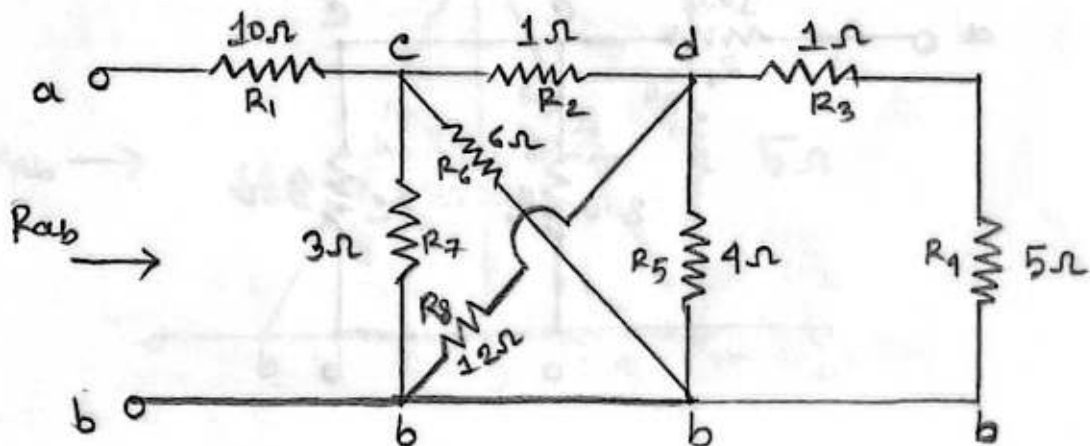
$$\therefore R_3 = R_3 + R_5 = 2 + 2 = 4\Omega$$

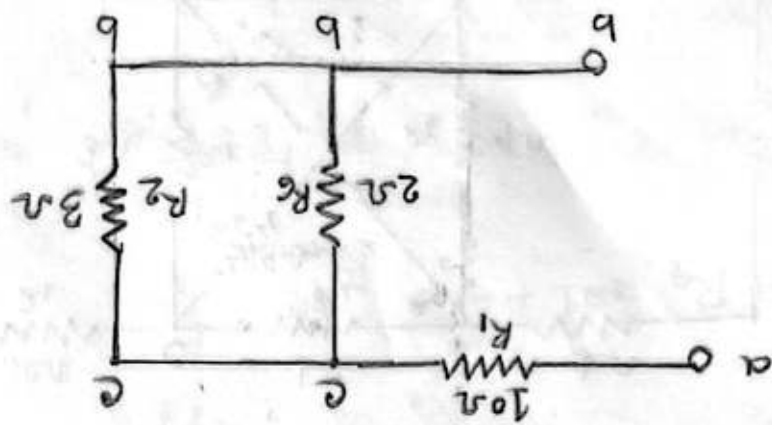
$$\therefore R_4 = \frac{R_3 \times R_4}{R_3 + R_4} = \frac{4 \times 6}{4 + 6} = 2.4\Omega$$



$$\therefore R_{eq} = 4 + 2.4 + 8 = 14.4\Omega \quad (\text{Ans.})$$

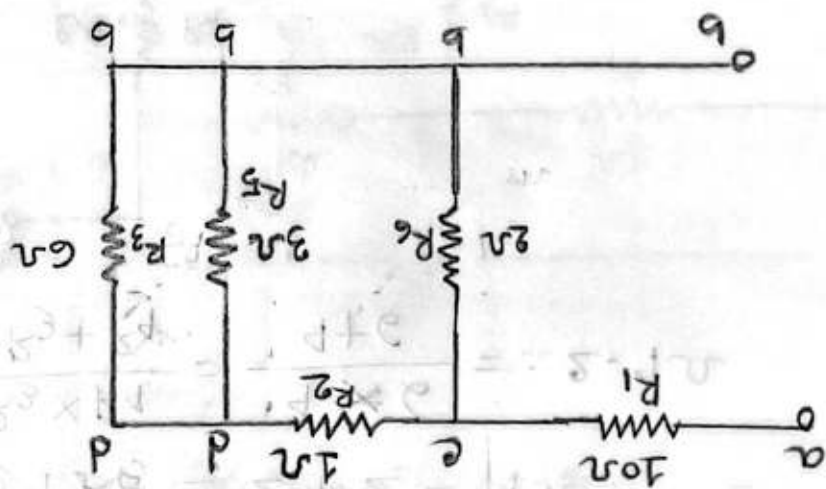
2.10





$$\therefore R_2 = 1 + 2 = 3\Omega$$

$$\therefore R_5 = \frac{6 \times 3}{6 + 3} = \frac{18}{9} = 2\Omega$$



$$\therefore R_5 = \frac{12 \times 4}{12 + 4} = \frac{48}{16} = 3\Omega$$

$$\therefore R_6 = \frac{6 \times 3}{6 + 3} = \frac{18}{9} = 2\Omega$$

$$\therefore R_3 = 1 + 5 = 6\Omega$$

Here,

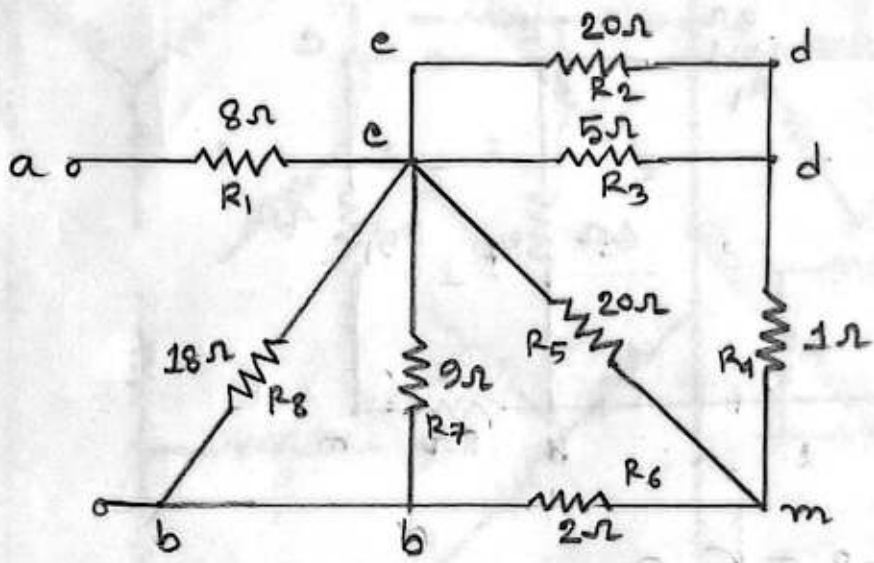
23.3.88

23.2.26

$$\therefore R_6 = \frac{3 \times 2}{2+3} = \frac{6}{5} = 1.2 \Omega$$

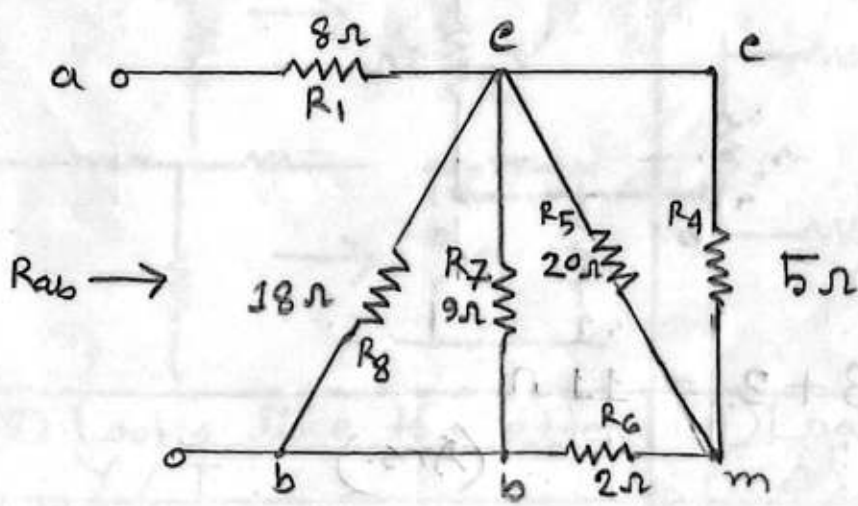
$$\therefore R_{ab} = 10 + 1.2 = 11.2 \Omega \quad (\text{Ans.})$$

P.2.10



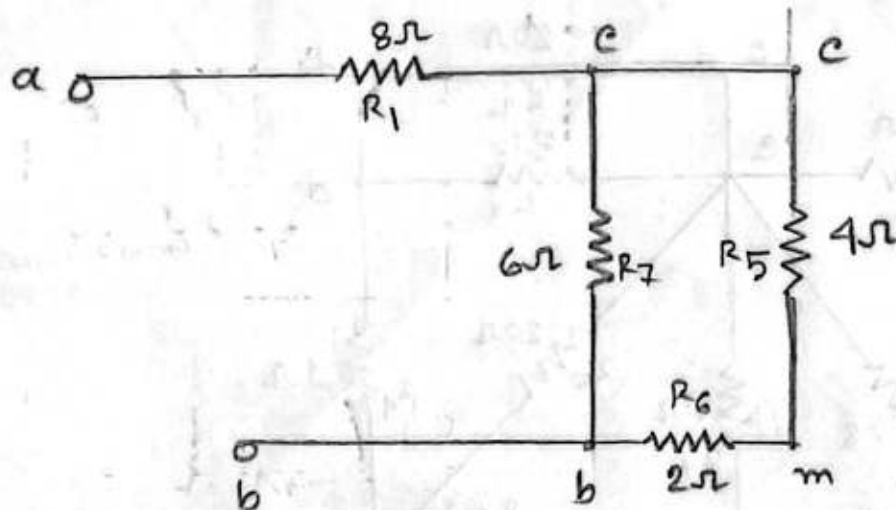
$$\therefore R_2 = \frac{20 \times 5}{20+5} = \frac{100}{25} = 4 \Omega$$

$$\therefore R_4 = 4 + 1 = 5 \Omega$$



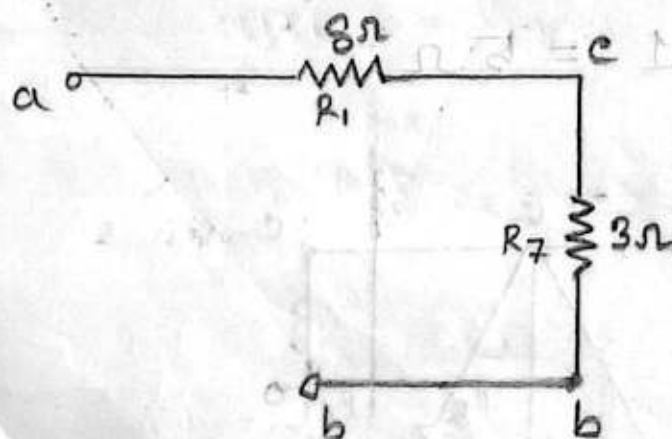
$$\therefore R_5 = \frac{20 \times 5}{20 + 5} = \frac{100}{25} = 4 \Omega$$

$$\therefore R_7 = \frac{18 \times 9}{18 + 9} = \frac{162}{27} = 6 \Omega$$



$$\therefore R_6 = 4 + 2 = 6 \Omega$$

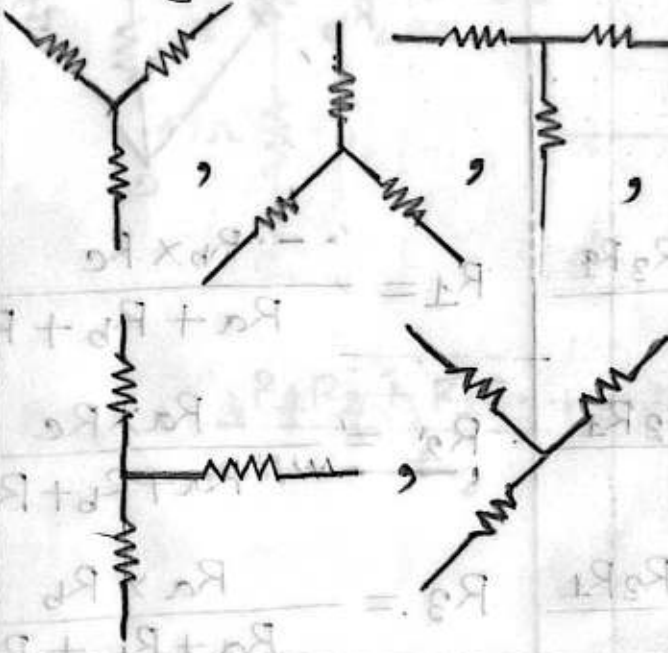
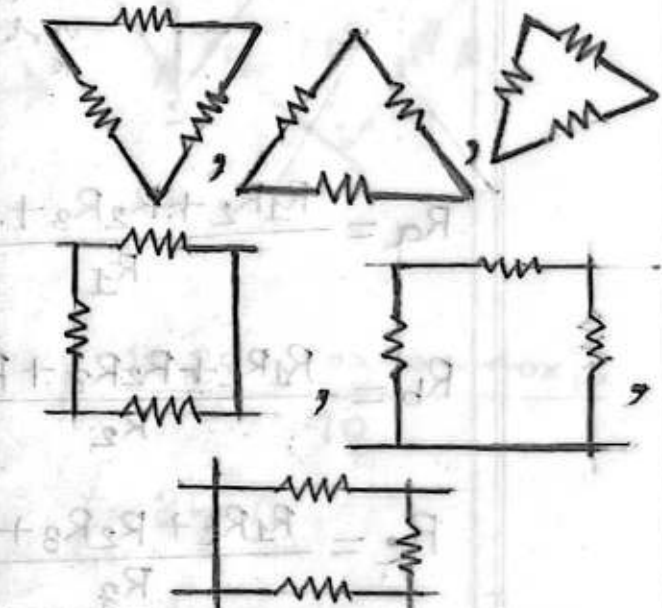
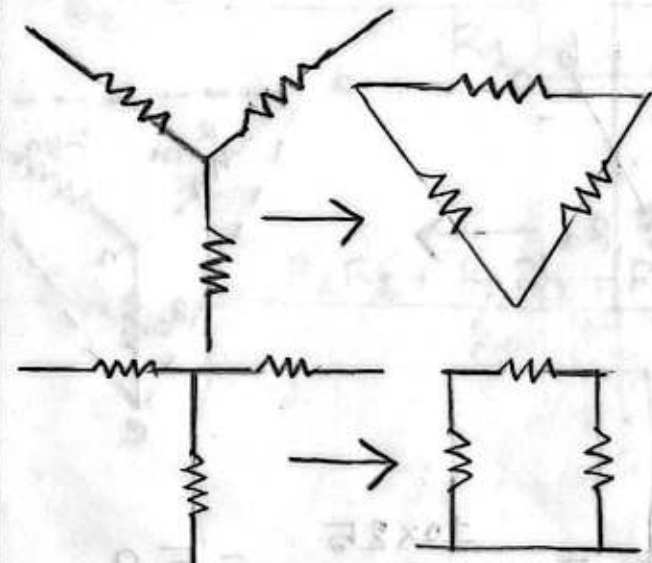
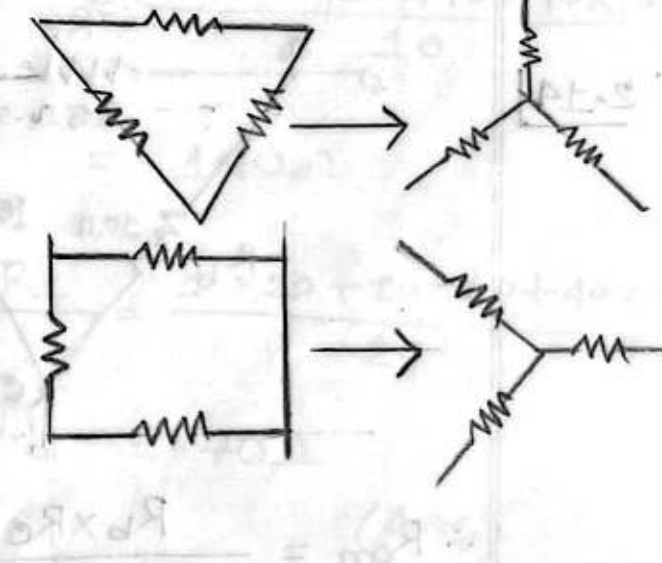
$$\therefore R_7 = \frac{6 \times 6}{6 + 6} = \frac{36}{12} = 3 \Omega$$

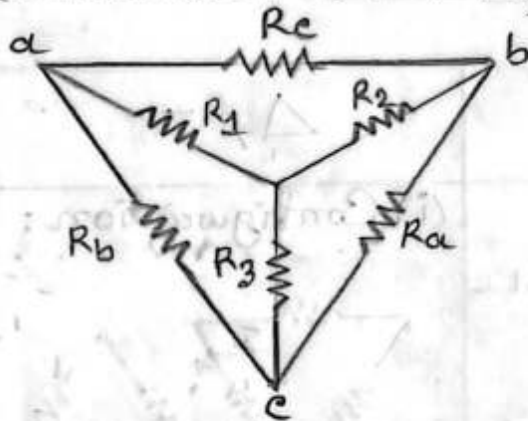


$$\therefore R_{ab} = 8 + 3 = 11 \Omega$$

(Ans.)

WYE-Delta Transformation -

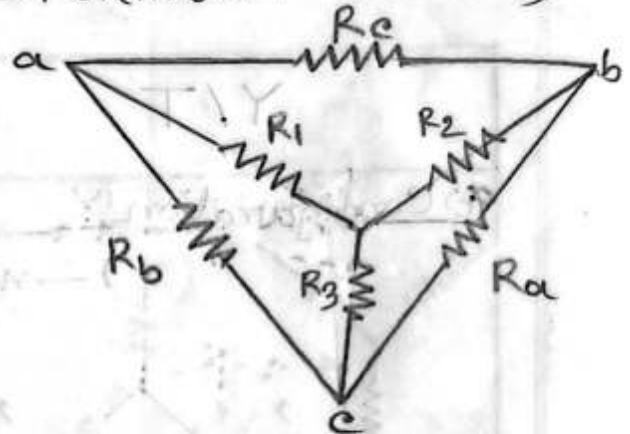
Y/T	Δ/π
(i) Configuration : 	(i) Configuration : 
(ii) Conversion : 	(ii) Conversion : 
(iii) Looks like the letter Y/T.	(iii) Looks like a triangle (Δ)/π.
(iv) Three resistors connected to a common point.	(iv) Three resistors connected in a closed loop.

(v) Formula: ($Y \rightarrow \Delta$)

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

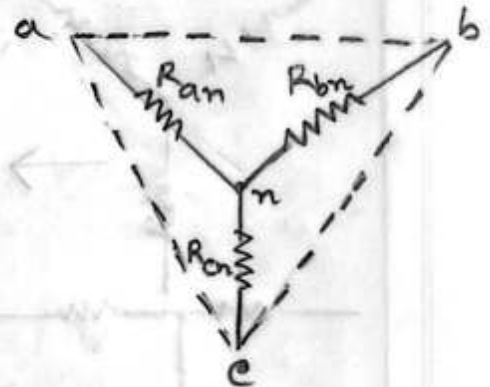
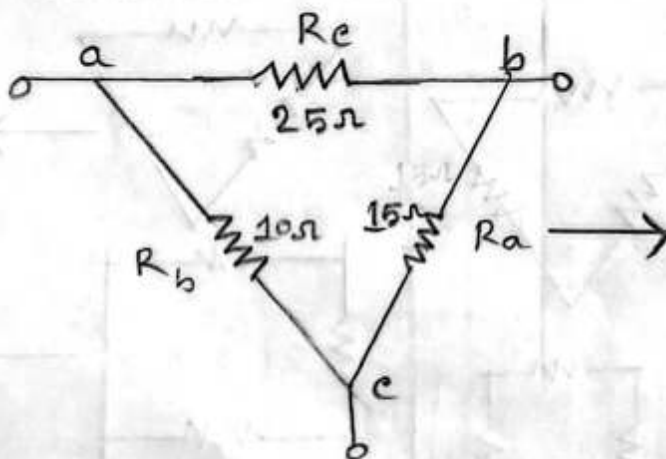
(v) Formula: ($\Delta \rightarrow Y$)

$$R_1 = \frac{R_b \times R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a \times R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a \times R_b}{R_a + R_b + R_c}$$

2.14

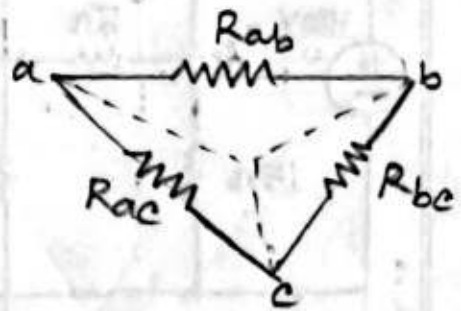
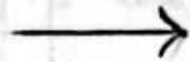
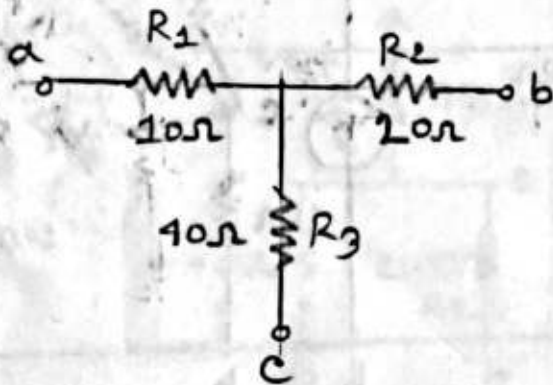


$$\therefore R_{cn} = \frac{R_b \times R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{10 + 15 + 25} = 5 \Omega$$

$$\therefore R_{bn} = \frac{R_a \times R_c}{R_a + R_b + R_c} = \frac{15 \times 25}{10 + 15 + 25} = 7.5 \Omega$$

$$\therefore R_{eq} = \frac{R_b \times R_c}{R_a + R_b + R_c} = \frac{10 \times 15}{10 + 15 + 25} = 3\Omega \quad (\text{Ans.})$$

P-2.14

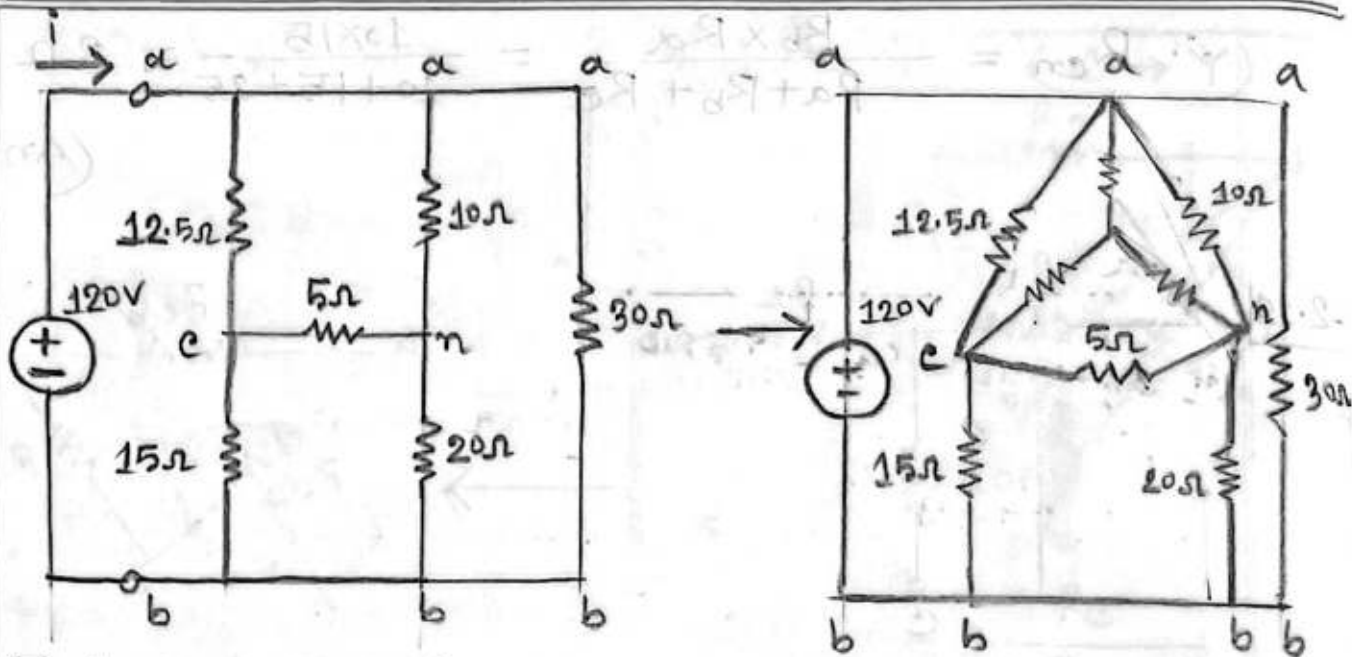
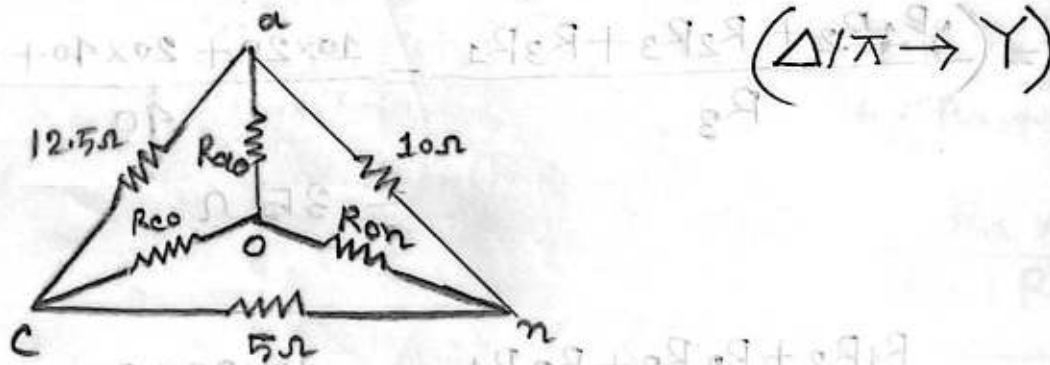


$$\therefore R_{ab} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} = \frac{10 \times 20 + 20 \times 40 + 40 \times 10}{40} = 35\Omega$$

$$\therefore R_{bc} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} = \frac{10 \times 20 + 20 \times 40 + 40 \times 10}{10} = 140\Omega$$

$$\therefore R_{ac} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} = \frac{10 \times 20 + 20 \times 40 + 40 \times 10}{20} = 70\Omega \quad (\text{Ans.})$$

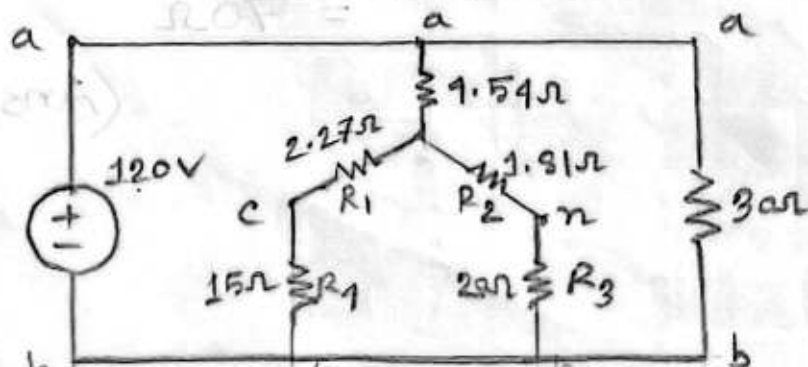
2.15

Find R_{ab} and i .

$$\therefore R_{ao} = \frac{12.5 \times 10}{12.5 + 10 + 5} = 4.54 \Omega$$

$$\therefore R_{co} = \frac{12.5 \times 5}{12.5 + 10 + 5} = 2.27 \Omega$$

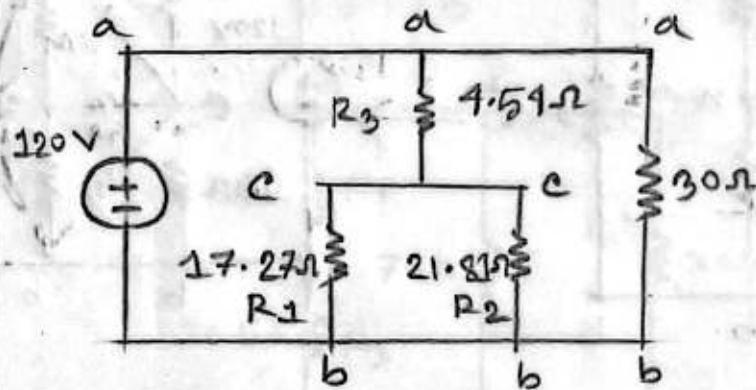
$$\therefore R_{on} = \frac{10 \times 5}{12.5 + 10 + 5} = 1.81 \Omega$$



23.2.26

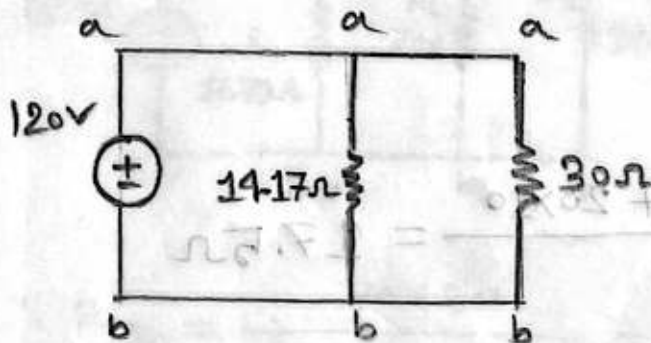
$$\therefore R_1 = 2.27 + 15 = 17.27 \Omega$$

$$\therefore R_2 = 1.81 + 20 = 21.81 \Omega$$



$$\therefore R_1 = \frac{17.27 \times 21.81}{17.27 + 21.81} = 9.63 \Omega$$

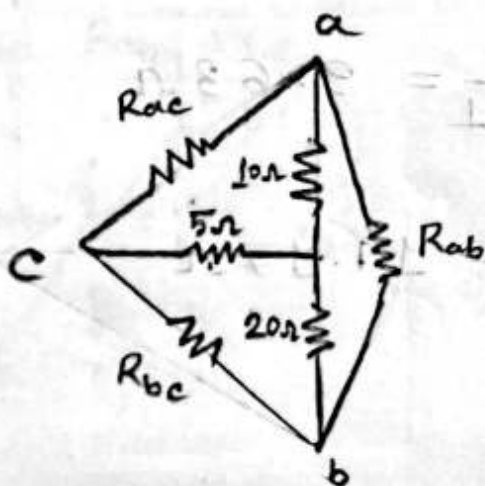
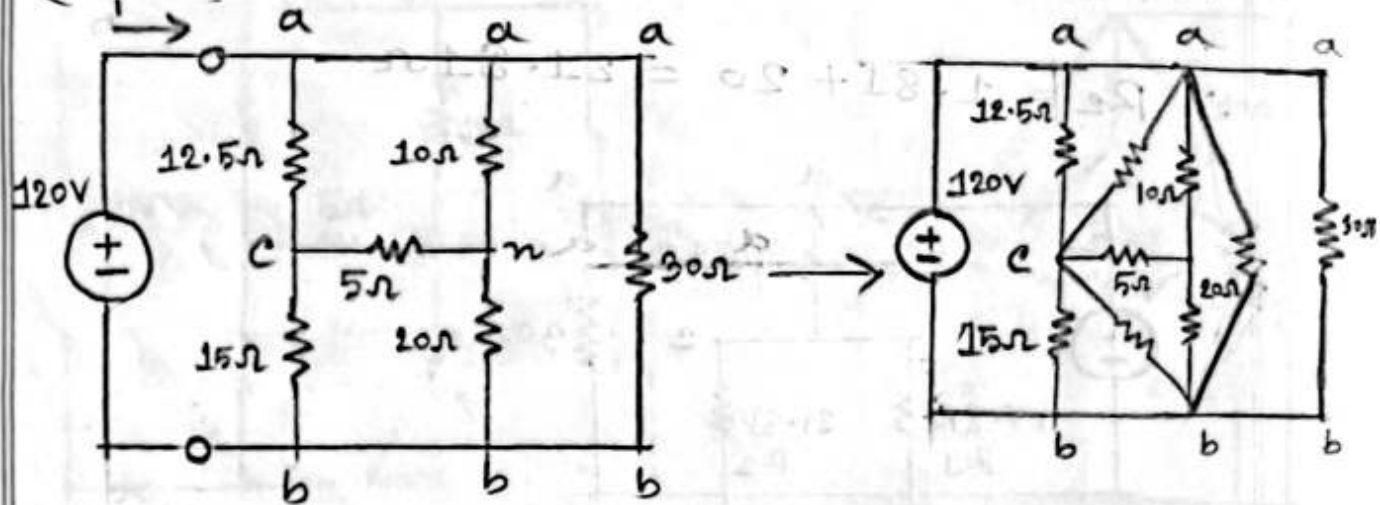
$$\therefore R_3 = 4.54 + 9.63 = 14.17 \Omega$$



$$\therefore R_{ab} = \frac{30 \times 14.17}{30 + 14.17} = 9.62 \Omega$$

$$\therefore i = \frac{V}{R_{ab}} = \frac{120}{9.62} = 12.47 \text{ A}$$

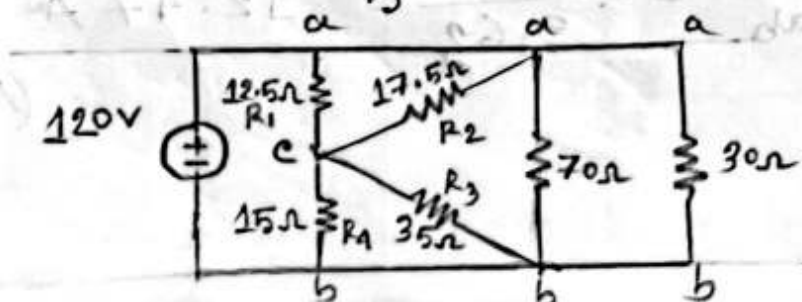
(Ans.)

(Y/T $\rightarrow$ Δ)

$$\therefore R_{ac} = \frac{10 \times 5 + 5 \times 20 + 20 \times 10}{20} = 17.5 \Omega$$

$$\therefore R_{bc} = \frac{10 \times 5 + 5 \times 20 + 20 \times 10}{10} = 35 \Omega$$

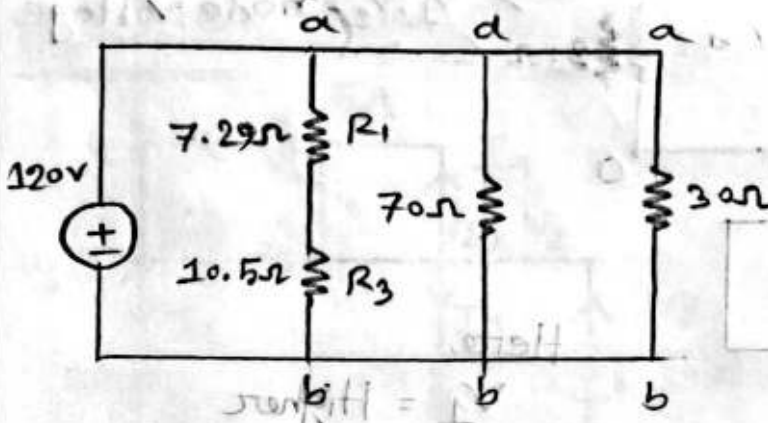
$$\therefore R_{ab} = \frac{10 \times 5 + 20 \times 5 + 20 \times 10}{5} = 70 \Omega$$



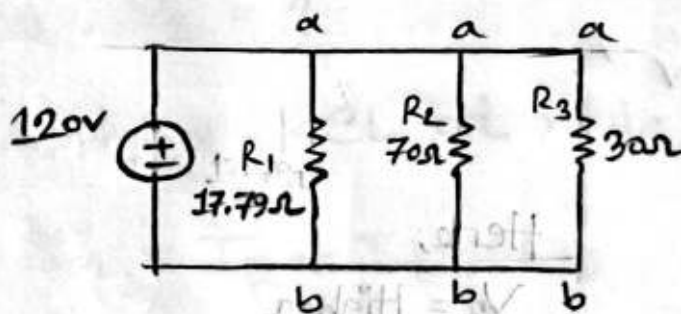
23.2.26

$$R_1 = \frac{12.5 \times 17.5}{12.5 + 17.5} = 7.29 \Omega$$

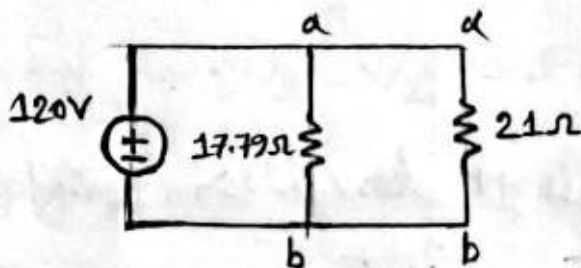
$$R_3 = \frac{35 \times 15}{35 + 15} = 10.5 \Omega$$



$$\therefore R_1 = 7.29 + 10.5 = 17.79 \Omega$$



$$\therefore R_{20} = \frac{70 \times 30}{70 + 30} = 21 \Omega$$



$$\therefore R_{ab} = \frac{21 \times 17.79}{17.79 + 21} = 9.62 \Omega$$

$$\therefore i = \frac{V}{R_{ab}} = \frac{120}{9.62} = 12.47 \text{ A (Ans.)}$$